

AI as a Tool

Analytics Team - Fall 2025

What is Dopamine??

Dopamine is a type of monoamine neurotransmitter

- **chemical messenger**
- **communicates between nerve cells in your brain and the rest of your body**

As humans, our brains are hard-wired to seek out behaviors that release dopamine in our reward system.



Quick History Recap

Our ancestors might have gotten dopamine hits from finding food, successful hunts, or meaningful social connections

They once got dopamine hits a few times a day—now we trigger dozens each hour from notifications, ordering food, or even reaching for your phone.

So what does this mean?

Our reward systems that have evolved over millions of years have been corrupted by artificial peaks our brains were never meant to handle.

The brain constantly tries to return to equilibrium.

The more often you spike dopamine, the more your baseline drops to compensate.

Anhedonia

an·hee·dow·nee·uh

"inability to feel pleasure in activities that are usually considered to be pleasurable."

The feeling of being both underwhelmed and overstimulated.

What does this have to do with chatgpting all my homework?

Using AI to complete assignments easily can give quick dopamine hits, but this quick stimulation can cause natural rewards (like completing a project or authentic learning) to feel inadequate over time.

As baseline dopamine drops, tolerance and withdrawal symptoms (such as boredom or decision fatigue) can develop, leading to diminished satisfaction from slow, effortful processes like creativity, social relationships, or healthy habits.

Cognitive Cost

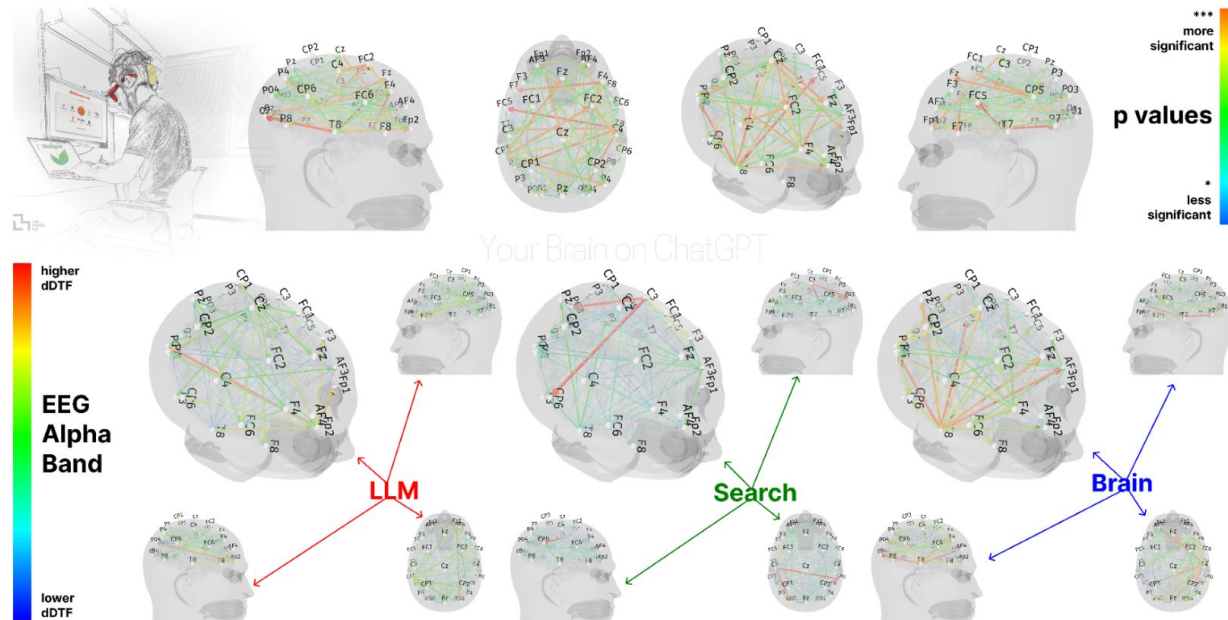


Figure 1. The dynamic Direct Transfer Function (dDTF) EEG analysis of Alpha Band for groups: LLM, Search Engine, Brain-only, including p-values to show significance from moderately significant (*) to highly significant (***).

AI Best Practices

Use AI and humans for what each does best:

AI: Repetitive, data-heavy tasks

Humans: Contextual and emotional understanding

AI Best Practices

Be More Present when Using AI:

Right now, most students use AI like a shortcut for essays, math problems, coding.

It gives answers instantly and hinders us from processing and retaining the information ourselves.

Instead of letting AI replace your brain, you can help it train your brain!

Think of AI as a free tutor that is available 24/7, for any subject, at any level and catered towards your learning style.

AI Best Practices

AI Tutor Prompts - Process

- Create a syllabus for the chosen subject or learning objectives.
- Ask me questions for each topic to assess my knowledge.
- Identify knowledge gaps and provide detailed explanations with external references (videos, articles, illustrations).
- Re-test me after each explanation to ensure mastery. Move to the next topic only after I've mastered the current one

AI Best Practices

AI Tutor Prompts - Example

I want to fully understand and apply **[insert session topic name]**.

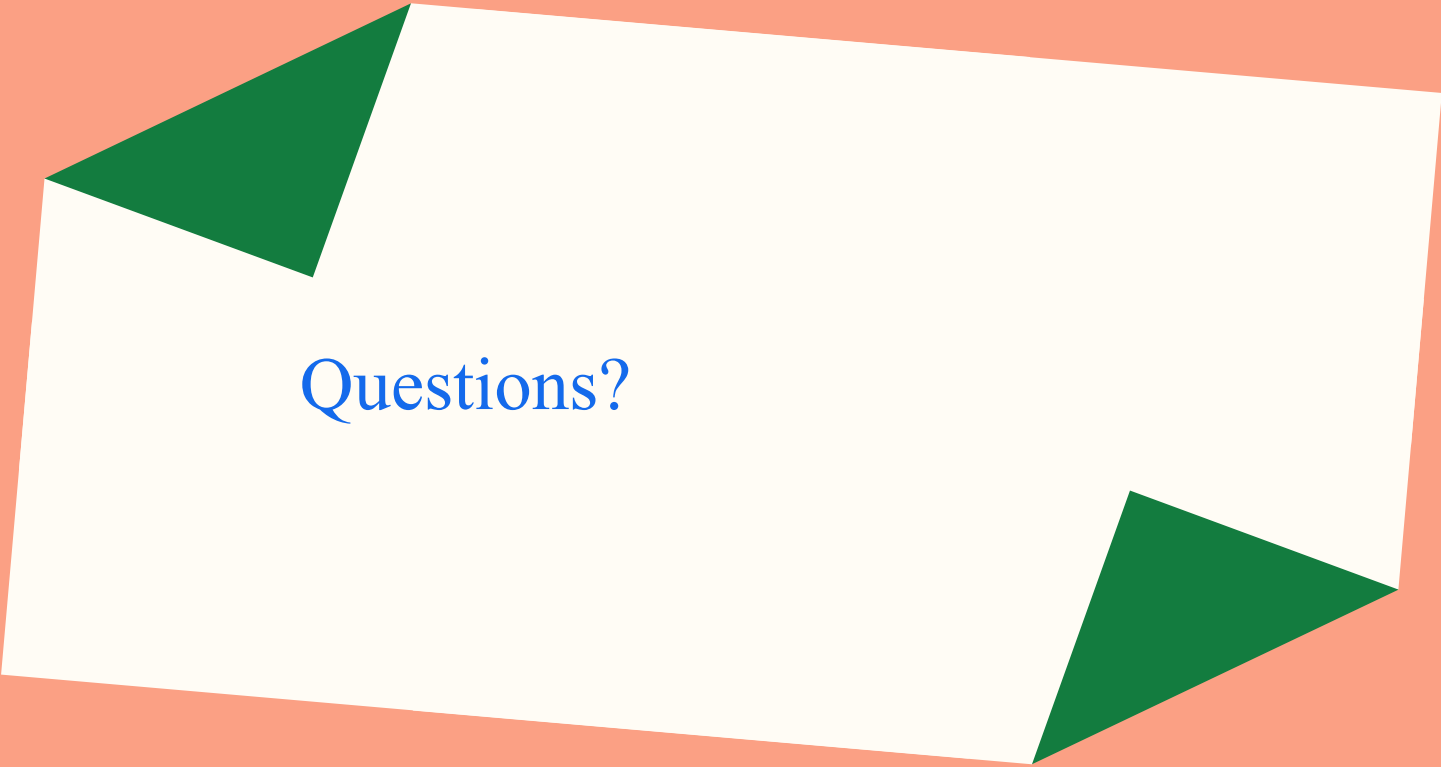
Act as my tutor and guide me through all 6 levels of mastery:

1. Know the definition
2. Calculate or explain it
3. Use it in a real scenario
4. Change something and predict the result
5. Teach it to someone else
6. Spot mistakes and correct them

Please don't give me direct answers, ask me questions in a Socratic way and help me reason things out.

Only give answers if I'm stuck and after you have provided a hint for me to work with and I still need support to move forward.

Also, ask questions one at a time so I can think through each step clearly.



Questions?